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TWO LAYER PIXEL STRUCTURE FOR HIGH RESOLUTION WITH HIGH DYNAMIC RANGE

Abstract

An image sensor achieves high pixel density, and therefore high resolution, by offloading portions of a photodetector circuit to a separate device layer from the photodiodes. The photodetector uses a lateral overflow integration capacitor and a dual conversion gain transistor to increase dynamic range. The dynamic range is further increased by providing a high conversion gain mode in which the floating diffusion node is isolated from the second device layer and from the wiring that extends to the second device layer. This is accomplished by disposing the DCG transistor and the source follower in the first device layer which has the photodiodes, the transfer gates, and the floating diffusion regions. Isolating the floating diffusion node from the wiring to the second device layer in the high conversion gain mode reduces the capacitance of the floating diffusion node in the high conversion gain mode, and so increases the dynamic range.

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Background/Summary

REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Application No. 63/558,147, filed on Feb. 27, 2024, the contents of which are incorporated herein by reference in their entirety.

BACKGROUND

[0002] Many modern day electronic devices (e.g., digital cameras, optical imaging devices, etc.) comprise image sensors. An image sensor includes an array of photosensitive areas which are operative as transducers that convert light into electrical charges. Examples of image sensors include charge-coupled device (CCD) image sensors and complementary metal-oxide-semiconductor (CMOS) image sensors. Compared to CCD image sensors, CMOS image sensors are favored due to low power consumption, small size, fast data processing, a direct output of data, and low manufacturing cost.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. In accordance with standard industry practice, features are not drawn to scale. Moreover, the dimensions of various features within individual drawings may be arbitrarily increased or reduced relative to one-another to facilitate illustration or provide emphasis.

[0004] FIG. 1 illustrates a three-layer device in accordance with some embodiments.

[0005] FIG. 2 provides a circuit diagram for an image sensor according to a first embodiment.

[0006] FIG. 3 illustrates a plan view of the first device layer in an image sensor in accordance with the first embodiment.

[0007] FIG. 4 provides a perspective view corresponding to FIG. 3.

[0008] FIG. 5 illustrates a plan view of the second device layer in an image sensor in accordance with the first embodiment.

[0009] FIG. 6 provides a perspective view corresponding to FIG. 5.

[0010] FIG. 7 provide a perspective view showing the first device layer and the second device layer together in accordance with the first embodiment.

[0011] FIG. 8 provides a circuit diagram for an image sensor according to a second embodiment.

[0012] FIG. 9 illustrates a plan view of the first device layer in an image sensor in accordance with the second embodiment.

[0013] FIG. 10 illustrates a plan view of the second device layer in an image sensor in accordance with the second embodiment.

[0014] FIG. 11 provides a circuit diagram for an image sensor according to a third embodiment.

[0015] FIG. 12 illustrates a plan view of the first device layer in an image sensor in accordance with the third embodiment.

[0016] FIG. 13 illustrates a plan view of the second device layer in an image sensor in accordance with the third embodiment.

[0017] FIGS. 14-21 provide a series of cross-sectional views illustrating an embodiment of a manufacturing process applied to the first device layer.

[0018] FIG. 22 provides a plan view corresponding to FIG. 21.

[0019] FIGS. 23-32 provide a series of cross-sectional views illustrating an embodiment of a manufacturing process applied to the second device layer.

[0020] FIG. 33 provides a plan view corresponding to FIG. 32.

[0021] FIGS. 34A-39 provide a series of cross-sectional views illustrating the joining of the first, second and third device layers and further manufacturing applied to the first device layer in accordance with an embodiment.

[0022] FIG. 40 is a flow chart of a process in accordance with some embodiments.

DETAILED DESCRIPTION

[0023] The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.

[0024] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper”, and the like, may be used herein to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. These spatially relative terms are intended to encompass different orientations of the device or apparatus in use or operation in addition to the orientation depicted in the figures. The device or apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may be interpreted accordingly. Terms “first”, “second”, “third”, “fourth”, and the like are merely generic identifiers and, as such, may be interchanged in various embodiments. For example, while an element (e.g., an opening) may be referred to as a “first” element in some embodiments, the element may be referred to as a “second” element in other embodiments.

[0025] One type of CMOS image sensor has an array of photodetectors each of which includes at least one photosensitive area within a semiconductor substrate, a transfer gate, a floating diffusion node, a source follower, a row select transistor, and a reset transistor. When the reset transistor is closed, the floating diffusion node is charged to a reference voltage. Light is transduced into electrical charges within the photosensitive area(s). The charges accumulate until the transfer gate is closed allowing them to flow to the floating diffusion node. The charges alter the floating diffusion node voltage. The floating diffusion node is coupled to a gate electrode of the source follower. The source follower is connected in series with the row select transistor. When the row select transistor is closed, current flows through the source follower and the row select transistor. The magnitude of that current depends on the floating diffusion node voltage that is applied to the source follower gate electrode. The current is detected and used to infer the amount of charge that was transferred to the floating diffusion node, which in turn reflects the amount of radiation that was incident on the photosensitive area(s) over a sampling interval.

[0026] Conversion gain is a significant parameter in a CMOS image sensor of the type just described. The conversion gain is related to the capacitance of the floating diffusion node. The capacitance of the floating diffusion node includes contributions from a floating diffusion region, which is the drain region of the transfer gate, the source region of the reset transistor, the gate electrode of the source follower, and the wiring that connects these structures. If the capacitance of the floating diffusion node is too low (conversion gain too high) for a given lighting condition and sampling rate, the photodetector will experience saturation so that variations in light intensity at the high end will be lost. If the capacitance is too high (conversion gain too low), there will be excessive noise in the voltage signal and variations in light intensity at the low end will be lost.

[0027] One approach to widening the range of lighting conditions over which the photodetector can

be effective (increasing dynamic range) involves adding a dual conversion gain (DCG) transistor to the photodetector circuit. The DCG transistor allows the photodetector to be switched between a low conversion gain mode and a high conversion gain mode. In the low conversion gain mode, the DCG transistor is closed so that a capacitor (commonly referred to as a lateral overflow integration capacitor) is coupled to the floating diffusion node through the DCG transistor. In the high conversion gain mode, the DCG transistor is open so that the lateral overflow integration capacitor (LOFIC) is isolated from the floating diffusion node.

[0028] Another significant performance parameter for a CMOS image sensor is resolution. High resolution is achieved through high pixel density. The chip area occupied by the transistors in the photodetector circuit can limit pixel density. One approach to overcoming that limitation is a two- or three-device layer approach in which the photodetector transistors (with the exception of the transfer gate) and the LOFIC are disposed on a second die. A shortcoming of this approach is that the wiring that connects the floating diffusion node to the second die adds capacitance to the floating diffusion node even in the high conversion gain mode. That extra capacitance reduces conversion gain and so increases noise and lowers dynamic range.

[0029] In accordance with the present disclosure, the problem of providing high resolution with high dynamic range is solved using a two- or three-device layer system in which the floating diffusion node is isolated from the wiring to the second die in the high conversion gain mode. This is accomplished by disposing the DCG transistor and the source follower on the first die together with the photosensitive area, the transfer gate, and the floating diffusion region. Some or all of the remaining photodetector circuit components are disposed on the second die so that high pixel density may still be achieved.

[0030] In some embodiments, the DCG transistor, the source follower, and other photodetector circuit components are shared so that there is one DCG transistor and one source follower for every four transfer gates and photosensitive regions. This type of sharing allows pixel density to be increased. The sharing arrangement entails uniting the floating diffusion regions corresponding to the four transfer gates into one floating diffusion node. This would seem to increase the capacitance of the floating diffusion node, however, under very low light conditions, when high conversion gain is at a premium, the four photodetector pixels may be operated as one large pixel. That operation mode increases full well capacity and provides higher dynamic range.

[0031] An image sensor according to the present disclosure may have a very large difference in conversion gain between the low conversion gain mode and the high conversion gain mode. In some embodiments, the photodetector includes a second DCG transistor and a second LOFIC so that there may be an intermediate conversion gain mode between the low conversion gain mode and the high conversion gain mode. The intermediate conversion gain mode may be used to provide a smoother transition from the low conversion gain mode to the high conversion gain mode. Alternatively, the intermediate conversion gain mode may take the place of the high conversion gain mode, and the high conversion gain mode may become a very high conversion gain mode so that the dynamic range is further increased.

[0032] In some embodiments, the source follower and the dual conversion gain transistor are aligned in a row. The four photosensitive areas that share the source follower and the dual conversion gain transistor may be arranged two on each side of the row. In some embodiments, each pair of transfer gates share a floating diffusion region, which is an area of the substrate on the drain side of the respective transfer gate. The DCG transistor and the source follower may be isolated from the floating diffusion regions and the photosensitive areas by shallow trench isolation structures. Wiring associated with the floating diffusion node may be limited to four contact plugs and interconnecting wiring in a metallization layer (the M1 metallization layer) nearest the substrate. The four contact plugs may include one contact plug for each of the two floating diffusion region, one contact plug for the source follower gate electrode, and one contact plug for the DCG transistor source region. All of these contact plugs may be coupled with two short wiring

segments in the M1 metallization layer. Wiring in the M2 and higher metallization layers may be avoided. This structure limits the wiring connected to the floating diffusion node in the high conversion gain mode and thereby increases dynamic range.

[0033] The first DCG and the second DCG may be connected in series and the first LOFIC may be connected to the first DCG in parallel with the second DCG. Closing the first DCG adds the first LOFIC to the floating diffusion node. Closing the second DCG in addition to the first DCG brings the second LOFIC into the floating diffusion node. In some embodiments, the first LOFIC is on the first die and the second LOFIC is on the second die. This arrangement allows the first LOFIC and the second LOFIC to each have a large area. Both the first LOFIC and the second LOFIC may be three-dimensional metal-insulator-metal (3DMIM) capacitors.

[0034] In some embodiments, each photodetector includes two connections between the first die and the second die. The first connection may be between the DCG transistor on the first die and an LOFIC or a reset transistor on the second die. The second connection may be between the source follower on the first die and a row select transistor on the second die. In some embodiments, the first and second connections form first rows, each first row comprising alternating first and second connections extending across the photodetector array. The first and second connection are made through bond pads on the first die. In some embodiments, dummy contact pads are disposed in second rows interleaved with the first rows. The dummy contact pads facilitate manufacturing. In some embodiments Vdd rails extend parallel to the first rows between the first rows. These Vdd rail positions avoid inductive coupling. In some embodiments, LOFIC capacitors are disposed directly below the Vdd rails.

[0035] In some embodiments, the second die has corresponding rows of bond pads and dummy contact pads. In some embodiments, the second die has Vdd rails extending parallel to the first rows between the first rows. In some embodiments, the second die has LOFIC capacitors disposed directly above the Vdd rails so that the LOFIC capacitors are between the Vdd rails and the substrate of the second die.

[0036] In some embodiments, the image sensor includes a third die. The third die may be in a stacked arrangement with the first die and the second die. The third die may contain an application specific integrated circuit (ASIC). The ASIC uses data from the photodetector array and may contain components such as memory cells, logic circuits, and the like. Placing this additional circuitry on the third die preserves the image sensing area on the first die and leaves more area for in-pixel circuitry on the second die.

[0037] FIG. 1 illustrates a perspective view of an integrated circuit device **100** in accordance with some embodiments. The integrated circuit device **100** includes three device layers: a first wafer **101A** comprising first dies **101**, a second wafer **103A** comprising second dies **103**, and a third wafer **105A** comprising third dies **105**. The first dies **101** and the second dies **103** include photodetector components. The third dies **105** may provide application specific integrated circuit (ASIC). After dicing along scribe lines **109** and then packaging, the integrated circuit device **100** provides image sensors **107** each of which includes a first die **101**, a second die **102**, and a third die **105**.

[0038] FIG. 2 provides a circuit diagram **200** for a photodetector that may be one in an array of photodetectors in the image sensor **107**. The photodetector includes four photodiodes PD, four transfer gates TX, a DCG transistor, and a source follower SF on the first die **101**. The four photodiodes PD are coupled to a floating diffusion node FD through four respective transfer gates TX. A gate of the source follower SF and the source side of the DCG transistor are also connected to the floating diffusion node FD.

[0039] The photodetector further includes a reset transistor RST, a row select transistor RSL, and a LOFIC on the second die **103**. The source side of the reset transistor RST is coupled to the drain side of the DCG transistor through a first connection structure **201A**. Closing the DCG transistor adds the capacitance of the first connection structure **201A** and related wiring as well as the

capacitance of the LOFIC to the floating diffusion node FD. The drain side of the row select transistor RSL is coupled to the source side of the source follower SF through a second connection structure **201B**. The source side of the row select transistor RSL may be coupled to an ASIC on the third die **105**. The ASIC is not considered part of the photodetector.

[0040] FIG. **3** provides a plan view **300** and FIG. **4** provides a perspective view **400** of an area on the first die **101** containing a portion of a photodetector **351** that is on the first die **101**. The photodetector **351** corresponds with the circuit diagram **200** of FIG. **2**. Referring to the plan view **300** of FIG. **3**, the photodetector **351** includes photodiodes **311A-D** in the first die **101**. The photodiodes **311A-D** are photosensitive areas in a semiconductor substrate **317**, which is part of the first die **101**. The photodiodes **311A** and **311B** are selectively couple to a first floating diffusion region **315A** through a first pair of transfer gates **319**. The photodiodes **311C** and **311D** are selectively coupled to a second floating diffusion region **315B** through a second pair of transfer gates **319**. The first floating diffusion region **315A** and the second floating diffusion region **315B** are doped areas of the semiconductor substrate **317**.

[0041] A DCG transistor **301** and a source follower **329** are arranged end to end in a row **302**. The photodiodes **311A** and **311B** are on one side of the row **302**, and the photodiodes **311C** and **311D** are on the opposite side of the row **302**. The DCG transistor **301** and the source follower **329** are isolated from one another, from the floating diffusion regions **315A-B**, and from the photodiodes **311A-D** by segments of a shallow trench isolation structure **333**. The DCG transistor **301** includes a DCG source region **309**, a DCG drain region **305**, and a DCG gate electrode **307**. The source follower **329** includes a source follower source region **325**, a source follower drain region **321**, and a source follower gate electrode **323**.

[0042] Wiring in the first metallization layer over the semiconductor substrate **317** connects the first floating diffusion region **315A**, the second floating diffusion region **315B**, the source follower gate electrode **323**, and the DCG source region **309** to provide a floating diffusion node **310**. The wiring includes a straight wire **331** that connects the first floating diffusion region **315A**, the source follower gate electrode **323**, and the second floating diffusion region **315B**, and an L-shaped wire **313** that connects the straight wire **331** to the DCG source region **309**.

[0043] With reference to the perspective view **400** of FIG. **4**, a first column of metal **303** connects the DCG drain region **305** to a contact pad **401A**. The contact pad **401A** is part of the first connection structure **201A** (see FIG. **2**). A second column of metal **327** connects the source follower source region **325** to a contact pad **401B**, which is part of the second connection structure **201B**.

[0044] The perspective view **400** of FIG. **4** is also a cutaway view and thereby shows some of the structures within and on the back side **406** of the semiconductor substrate **317**. These include a back side deep trench isolation structure **413** that provides electrical isolation between adjacent photodiodes **311A-D**. A back side metal grid **411** increases optical isolation between adjacent photodiodes **311A-D**. Color filters **409** and microlenses **407** may also be disposed on the back side **406**.

[0045] FIG. **5** provides a plan view **500** and FIG. **6** provides a perspective view **600** of an area on the second die **103** that contains a second portion of the photodetector **351**. Referring to the plan view **500** of FIG. **5**, the second die **103** includes a semiconductor substrate **517**. A reset transistor **509** and a row select transistor **529** are formed on the semiconductor substrate **517** and are surrounded by isolations structures **505** and **501** respectively. The reset transistor **509** includes a reset transistor source region **513**, a reset transistor gate electrode **511**, and a reset transistor drain region **507**. The row select transistor **529** includes a row select transistor source region **521**, a row select transistor gate electrode **525**, and a row select transistor drain region **531**. An LOFIC **515** is disposed over the semiconductor substrate **517** and a V.sub.dd rail **503** is disposed over the LOFIC **515**.

[0046] With reference to the perspective view **600** of FIG. **6**, a first column of metal **603A** and a

wire **519** connect the reset transistor source region **513** and a bottom electrode **607** of the LOFIC **515** to a contact pad **601A**, which is part of the first connection structure **201A** (see FIG. 2). A second column of metal **603B** connects the row select transistor drain region **531** to a contact pad **601B**, which is part of the second connection structure **201B**.

[0047] FIG. 7 provides a perspective view **700** showing how the first die **101** is coupled to the second die **103**. In particular, the contact pad **401A** abuts the contact pad **601A** to form the first connection structure **201A** and the contact pad **401B** abuts the contact pad **601B** to form the second connection structure **201B**. It can also be seen from the perspective view **700** that the first column of metal **303**, the second column of metal **327**, and the connection structures **201A** and **201B** can be electrically isolated from the floating diffusion node **310**.

[0048] FIG. 8 provides a circuit diagram **800** for another photodetector that may be used in the image sensor **107**. The circuit diagram **800** is like the circuit diagram **200** of FIG. 2 except that in the circuit diagram **800** a second LOFIC (LOFIC2) is provided in the first die **101** and a second DCG transistor (DCG2) is provided in the second die **102**. Closing the DCG1 transistor adds the second LOFIC to the floating diffusion node FD and closing both the DCG1 transistor and the DCG2 transistor adds both the first LOFIC and the second LOFIC to the floating diffusion node FD.

[0049] FIG. 9 provides a plan view **900** of an area on the first die **101** containing that portion of a photodetector **951** which is on the first die **101**. The photodetector **951** corresponds with the circuit diagram **800** of FIG. 8. The portion of the photodetector **951** that is on the first die **101** is like the portion of the photodetector **351** that would be on the first die **101** (see FIG. 3) except that the photodetector **951** include a second LOFIC **905**. A wire **901** connects a bottom electrode (not shown) of the second LOFIC **905** to the first column of metal **303** and the DCG drain region **305**. A top electrode **904** of the second LOFIC **905** is connected to a V.sub.dd rail **903**, which is above the second LOFIC **905** in a metal interconnect structure of the first die **101**. The V.sub.dd rail **903** may be one in an array of V.sub.dd rails **903** interleaved with rows **302**.

[0050] FIG. 10 provides a plan view **1000** of an area on the second die **103** that contains that portion of the photodetector **951** which is on the second die **103**. The portion of the photodetector **951** that is on the second die **103** is like the portion of the photodetector **351** that would be on the second die **103** (see FIG. 5) except that the photodetector **951** includes a second DCG **1003** on the second die **103**. The second DCG **1003** includes a drain region **1001**, a source region **1007**, and a gate electrode **1005**. Instead of being coupled directly to the first connection structure **201A** as it is in the photodetector **351** (see FIG. 5), the wire **519** connects to the first connection structure **201A** through the second DCG **1003**.

[0051] FIG. 11 provides a circuit diagram **1100** for a photodetector according to another embodiment. The circuit diagram **1100** is like the circuit diagram **200** of FIG. 2 except that in the circuit diagram **1100** there is one less connection between the first die **101** and the second die **103** and the LOFIC and the reset transistor RST are on the first die **101**. Moving these components to the first die **101** leaves more area on the second die **103** for in-pixel circuitry. The additional area on the second die **103** might alternatively be used to implement the ASIC.

[0052] FIG. 12 provides a plan view **1200** of an area on the first die **101** containing a portion of a photodetector **1251** that is on the first die **101**. The photodetector **1251** corresponds with the circuit diagram **1100** of FIG. 11. The portion of the photodetector **1251** that is on the first die **101** is like the portion of the photodetector **351** that would be on the first die **101** (see FIG. 3) except that the photodetector **1251** include a reset transistor **1203** and an LOFIC **1209**. Moreover, the photodetector **1251** lacks the column of metal **303** (see FIG. 3) and instead contains a wire **1207** coupling the DCG drain region **305** to a source region **1205** of the reset transistor **1203** and to a bottom electrode of the LOFIC **1209**. The top electrode of the LOFIC **1209** is connected to a V.sub.dd rail (not shown).

[0053] FIG. 13 provides a plan view **1300** of an area on the second die **103** that contains a second

portion of the photodetector **1251** that is on the second die **103**. The portion of the photodetector **1251** that is on the second die **103** is like the portion of the photodetector **351** that would be on the second die **103** (see FIG. 5) except that the photodetector **951** lacks the reset transistor **509** and the LOFIC **515**.

[0054] FIGS. **14-39** provide a series of views that illustrate an image sensor according to the present disclosure at various stages of manufacture according to a process of the present disclosure. The cross-sectional views in this series of views may correspond with the line A-A' in FIGS. **3** and **5**. Although FIGS. **14-39** are described in relation to a series of acts, it will be appreciated that the order of the acts may in some cases be altered and that this series of acts are applicable to structures other than the ones illustrated. In some embodiments, some of these acts may be omitted in whole or in part. Furthermore, although FIGS. **14-39** are described in relation to a series of acts, it will be appreciated that the structures shown in FIGS. **14-39** are not limited to a method of manufacture but rather may stand alone as structures separate from the method.

[0055] As illustrated by the cross-sectional view **1400** of FIG. **14**, the method may begin with forming the shallow trench isolation structures **333** in the semiconductor substrate **317** of the first wafer **101A**. The semiconductor substrate **317** may be cut from a single crystal and may be any type of semiconductor. The semiconductor may be, for example, silicon (Si), a group III-V semiconductor or some other binary semiconductor, a tertiary semiconductor (e.g., AlGaAs), a higher order semiconductor, or the like. In some embodiments, the semiconductor substrate **317** is or comprises silicon (Si) or the like. Forming the shallow trench isolation structures **333** may include forming a mask and etching trenches in the semiconductor substrate **317**, stripping the mask, depositing a dielectric so as to fill the trenches, and planarizing. The dielectric may be silicon dioxide (SiO₂), the like, or any other suitable dielectric.

[0056] As illustrated by the cross-sectional view **1500** of FIG. **15**, the method may continue with implanting dopants to form the photodiodes **311A-D**. The dopants may be implanted in a series of steps that include, for example, a deep n-well implant, a shallow p-well implant, and the like. Some of these dopant implants may be carried out with masks and others without. Some of these dopant implants may be carried out before the shallow trench isolation structures **333** are formed.

[0057] As illustrated by the cross-sectional view **1600** of FIG. **16**, transistor gates including gate dielectrics **1601** and gate electrodes **1603** may be formed on the semiconductor substrate **317** in order to provide transfer gates **319** and other transistors. The gate dielectrics **1601** may be an oxide, the like, or some other suitable material. The gate electrodes **1603** may be polysilicon, the like, or some other suitable material. These layers may be deposited by physical vapor deposition (PVD), chemical vapor deposition (CVD), atomic layer deposition (ALD), the like, or any other suitable process. The gate dielectrics **1601** may be formed by oxidation. After deposition, these layers are patterned to define individual gates. Spacers (not shown) are typically formed around the gates. Spacers may be formed by depositing a spacer material followed by anisotropic etching. The spacer material may include one or more layers of any suitable dielectrics. The spacer material may be or comprise, for example, silicon nitride (SiN), silicon oxynitride (SiON), silicon dioxide (SiO₂), a high-K dielectric, or the like. The spacer material may be deposited by ALD, CVD, PVD, the like, or any other suitable process.

[0058] As shown by the cross-sectional view **1700** of FIG. **17**, dopants may be implanted to form floating diffusion regions **315A-B**, the source follower source region **325**, and other source/drain regions. A mask **1701** may be formed prior to implanting the dopants. The floating diffusion regions **315A-B** may be aligned to the transfer gates **319** and to the shallow trench isolation structures **333**. Alternatively, the floating diffusion regions **315A-B** may be spaced apart from the shallow trench isolation structures **333** to reduce leakage currents.

[0059] As shown by the cross-sectional view **1800** of FIG. **18**, the process may continue with formation of a metal interconnect structure **371**. The metal interconnect structure **371** includes a plurality of metallization layers, contact plugs **1801**, vias **1809**, and a plurality of metallization

layers that include wires **1803**. The metallization layers include an M1 metallization layer, which is closest to the semiconductor substrate **317**, an M2 metallization layer, and an M.sub.top metallization layer, which is uppermost. A greater or lesser number of metallization layers may be included. V.sub.dd rails **373** may be disposed in the M.sub.top metallization layer or some other metallization layer.

[0060] The contact plugs **1801** may be or comprise tungsten (W), cobalt (Co), cobalt silicide (CoSi₂), nickel (Ni), nickel silicide (NiSi), an alloy thereof, the like, or some other suitable material. The wires **1803** and the vias **1809** may be or comprise copper (Cu), tungsten (W), ruthenium (Ru), palladium (Pd), platinum (Pt), cobalt (Co), nickel (Ni), zirconium (Zi), titanium (Ti), tantalum (Ta), aluminum (Al), conductive carbides, oxides, alloys of these metals, the like, or any other suitable conductive materials. The wires **1803** and the vias **1809** may also include a diffusion barrier layer such as titanium (Ti), tantalum (Ta), titanium nitride (TiN), tantalum nitride (Ta₂N), or the like. The metallization layers may be formed by damascene or dual damascene processes. The conductive materials in the metal interconnect structure **371** may be deposited by electroplating, electroless plating, ALD, CVD, PVD, the like, or any other suitable processes.

[0061] An interlevel dielectric **1807** surrounds the wires **1803** and the vias **1809**. Adjacent metallization layers may be separated by etch stop layers **1805**. The interlevel dielectric **1807** may include one or more layers of silicon dioxide (SiO₂), a low-κ dielectric, or an extremely low-κ dielectric. A low-κ dielectric is one having a smaller dielectric constant than silicon dioxide (SiO₂). Silicon dioxide has a dielectric constant of about 3.9. Examples of low-κ dielectrics include organosilicate glasses (OSG) such as carbon-doped silicon dioxide, fluorine-doped silicon dioxide (otherwise referred to as fluorinated silica glass (FSG)), organic polymer low low-κ dielectrics, and porous silicate glass. An extremely low-κ dielectric is a material having a dielectric constant of about 2.1 or less. An extremely low-κ dielectric material is generally a low-κ dielectric material with a porous structure. Porosity reduces the effective dielectric constant. The etch stop layers **1805** may include one or more layers of silicon nitride (SiN), silicon carbide (SiC), silicon carbonitride (SiCN), silicon oxycarbide (SiOC), silicon oxycarbonitride (SiOCN), combinations thereof, or the like. The interlevel dielectric **1807** and the etch stop layers **1805** may be deposited by ALD, CVD, PVD, the like, or any other suitable processes.

[0062] As shown by the cross-sectional views **1900-2100** of FIGS. **19-21**, a bonding layer **377** may be formed over the metal interconnect structure **371**. As shown by the cross-sectional view **1900** of FIG. **19**, formation of the bonding layer **377** may begin with deposition and patterning of an etch stop layer **1901** and a passivation layer **1903**. The passivation layer **1903** may include one or more layers of materials such as silicon dioxide (SiO₂), silicon nitride (SiN), the like, or other suitable dielectrics. Openings **1905** and **1907** may be formed by masking and etching.

[0063] As shown by the cross-sectional view **2000** of FIG. **20**, a metal **2001** may be deposited so as to fill the openings **1905** and **1907**. The metal **2001** may be or comprise copper (Cu), aluminum (Al), silver (Ag), tin (Sn), Indium (In), nickel (Ni), conductive carbides, oxides, alloys of these metals, the like, or any other suitable materials. The metal **2001** may be deposited by electroplating, electroless plating, ALD, CVD, PVD, the like, or any other suitable processes.

[0064] As shown by the cross-sectional view **2100** of FIG. **21** a planarization process may be carried out to remove the metal **2001** that deposited outside the openings **1905** and **1907**. The metal **2001** that remains in the openings **1905** provides contact pads **401B** and the like. The metal **2001** that remains in the openings **1907** provides dummy contact pads **375**. The planarization process may be chemical mechanical polishing (CMP), the like, or some other suitable process.

[0065] FIG. **22** provides a plan view **2200** corresponding to FIG. **21**. As shown by the plan view **2200**, the dummy contact pads **375** form rows **2201** between the rows **302** that include the contact pads **401A** and **401B**. The dummy contact pads **375** combine with the contact pads **401A** and **401B** to form a regular grid so that the dummy contact pads **375** prevent dishing during the CMP process illustrated by the cross-sectional view **2100** of FIG. **21**. The plan view **2200** also shows that the

V.sub.dd rails **373** run parallel to the rows **302** and are disposed between the rows **302**. The extensive coverage of the V.sub.dd rails **373** prevents voltage drops that could cause noise. The positioning of the V.sub.dd rails **373** avoids noise related to inductive coupling with the wires that connect to the contact pads **401A** and **401B**.

[0066] The cross-sectional views **2300** to **3200** of FIGS. **23** to **32** illustrate processing applied to make the second wafer **103A**. As shown by the cross-sectional view **2300** of FIG. **23**, the processing may begin with forming isolation structures **501** in the semiconductor substrate **517**. The semiconductor substrate **517** may have one of the compositions suitable for the semiconductor substrate **317** (see FIG. **14**). Transistors may be formed on the semiconductor substrate **517** as they are on the semiconductor substrate **317** (see FIGS. **16** and **17**), although none are shown in the cross-sectional views **2300** to **3200** of FIGS. **23** to **32**.

[0067] As shown by the cross-sectional view **2400** of FIG. **24**, a first group of metallization layers in the metal interconnect structure **575** may be formed over the semiconductor substrate **517**. The compositions of these layers and their methods of formation may be as described for the metal interconnect structure **371** in connection with FIG. **18**.

[0068] The cross-sectional views **2500** to **3000** of FIGS. **23** to **32** illustrate an example process of forming the LOFIC **515** (see FIG. **6**). As shown by the cross-sectional view **2500** of FIG. **25**, the process may begin with deposition of an etch stop layer **2501**, an interlevel dielectric layer **2503** and a mask **2505**. The mask **2505** is patterned and used to etch one or more trenches **2507**. A wire **2509** that will be used for a bottom electrode contact may be exposed by the trenches **2507**.

[0069] As shown by the cross-sectional view **2600** of FIG. **26**, a bottom electrode layer **2603** may be deposited so as to line the trenches **2507** and contact the wire **2509**. The bottom electrode layer **2603** may be, for example, titanium nitride (TiN), tungsten (W), titanium (Ti), tantalum (Ta), titanium nitride (Ta_N), copper (Cu), silver (Ag), aluminum (Al), nickel (Ni), a conductive alloy thereof, or the like. In some embodiments the bottom is titanium nitride (TiN) or the like. Using titanium nitride (TiN) for the electrode metal layer contributes to exceptionally low equivalent series resistance. In some embodiments, the bottom electrode layer **2603** is deposited to a thickness in the range from about 1 nm to about 20 nm. In some embodiments, the bottom electrode layer **2603** is deposited to a thickness in the range from about 20 nm to about 50 nm. Thinner electrode metal layers allow more capacitor plates to be deposited in the trenches **2507** and can provide higher capacitance. Thicker electrode metal layers can reduce equivalent series resistance. The bottom electrode layer **2603** may be deposited by physical vapor deposition (PVD), chemical vapor deposition (CVD), atomic layer deposition (ALD), electroplating, electroless plating, the like, or any other suitable process.

[0070] A sacrificial layer **2601** may be deposited over the bottom electrode layer **2603** so as to fill the trenches **2507**. The sacrificial layer **2601** may be a bottom antireflective coating (BARC) or any other suitable material. As shown by the cross-sectional view **2700** of FIG. **27**, the sacrificial layer **2601** facilitates a planarization process. The planarization process may be CMP or the like and removes portions of the bottom electrode layer **2603** that deposited outside the trenches **2507**. A remaining portion of the bottom electrode layer **2603** provides the bottom electrode **607**. As shown by the cross-sectional view **2800** of FIG. **28**, after planarization the sacrificial layer **2601** may be removed.

[0071] As shown by the cross-sectional view **2900** of FIG. **29**, a capacitor dielectric layer **606** and the top electrode **605** may be deposited over the structure shown by the cross-sectional view **2800** of FIG. **28**. The capacitor dielectric layer **606** may be any suitable dielectric. In some embodiments, the capacitor dielectric layer **606** is a high-k dielectric. Examples of high-k dielectrics include, without limitation, hafnium oxide (HfO₂), hafnium silicon oxide (HfSiO), hafnium silicon oxynitride (HfSiON), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), hafnium zirconium oxide (HfZrO), hafnium oxide aluminum oxide (HfO₂—Al₂O₃), zirconium oxide (ZrO₂), tantalum oxide (Ta₂O₅), aluminum oxide

(Al.sub.2O.sub.3), yttrium oxide (Y.sub.2O.sub.3), lanthanum oxide (La.sub.2O.sub.3), strontium titanium oxide (SrTiO.sub.3), and the like. The capacitor dielectric layer **606** may be deposited to any suitable thickness. In some embodiments, the capacitor dielectric layer **606** is deposited so as to have a thickness in the range from about 5 nm to about 20 nm. The capacitor dielectric layer **606** may be deposited by ALD, CVD, PVD, the like, or any suitable process. The top electrode **605** may have a composition and thickness as described for the bottom electrode **607**.

[0072] The cross-sectional view **2900** shows the trenches **2507** being substantially filled by the top electrode **605**. Alternatively, the trench sizes and layer thickness may be selected so that additional dielectric layers and electrode layers may be deposited prior to filling the trenches **2507** so that the resulting capacitor may have a greater number of plates than the number illustrated. As the trenches **2507** fill, inclusions **2901** may be formed within the trenches **2507**.

[0073] As shown by the cross-sectional view **3000** of FIG. **30**, a mask **3001** may be formed and used to pattern the capacitor dielectric layer **606** and the top electrode **605** so as to complete the formation of the LOFIC **515**. The LOFIC **515** is a trench-type capacitor, which may also be described as a 3D-MIM capacitor. More generally, the LOFIC **515** may be any suitable type of capacitor.

[0074] As shown by the cross-sectional view **3100** of FIG. **31**, after formation of the LOFIC **515** the formation of the metal interconnect structure **575** may be completed. The metal interconnect structure **575** may include the V.sub.dd rail **503**. In some embodiments the V.sub.dd rail **503** is directly over the LOFIC **515**.

[0075] As shown by the cross-sectional view **3200** of FIG. **32**, a bonding layer **573** is formed over the metal interconnect structure **575**. The bonding layer **573** includes the contact pad **601B** and dummy contact pads **571**. The formation and composition of the bonding layer **573** may be as described for the bonding layer **377** in connection with FIGS. **19-21**.

[0076] FIG. **33** provides a plan view **3300** corresponding to FIG. **32**. As shown by the plan view **3300**, the dummy contact pads **571** form rows **3301** between the rows **3303** that contains the contact pads **601A** and **601B**. The dummy contact pads **571** combine with the contact pads **601A** and **601B** to form a regular grid so that the dummy contact pads **571** prevent dishing during the CMP. As shown in the plan view **3300**, the V.sub.dd rails **503** run parallel to the rows **3303** and are disposed between the rows **3303**. The positioning of the V.sub.dd rails **503** avoids noise related to inductive coupling with the wires that connect to the contact pads **601A** and **601B**.

[0077] As shown by the cross-sectional view **3400** of FIG. **34A**, the first wafer **101A** and the second wafer **103A** are aligned and bonded together. The bonding may be metal-to-metal bonding, dielectric-to-dielectric bonding, or both metal-to-metal and dielectric-to-dielectric (hybrid) bonding. The contact pad **401B** is brought into abutment with the contact pad **601B** to form the second connection structure **201B**.

[0078] As shown by the cross-sectional view **3410** of FIG. **34B**, the semiconductor substrate **517** of the second wafer **103A** may be thinned from the back side **3409**. The semiconductor substrate **517** may be thinned by etching, mechanical grinding, CMP, the like, or any other suitable process. After thinning, through substrate vias (TSVs, not shown) may be formed. The TSVs may connect contact pads **3401**, which are formed on the back side **3409**, to structures on the front side **3407** such as the metal interconnect structure **575** and the row select transistor source region **521** (see FIG. **5**).

[0079] As shown by the cross-sectional view **3420** of FIG. **34C**, the third wafer **105A** and the second wafer **103A** are aligned and bonded together. The bonding may be metal-to-metal bonding, dielectric-to-dielectric bonding, or both metal-to-metal and dielectric-to-dielectric bonding. Contact pads **3403** on the third wafer **105A** may form connections with contact pads **3401** on the second wafer **103A**.

[0080] As shown by the cross-sectional view **3500** of FIG. **35**, the semiconductor substrate **317** may be thinned from the back side **406**. Thinning the semiconductor substrate **317** allows light to pass more easily to the photodiodes **311A-D** (see FIG. **3**). The semiconductor substrate **317** may be

thinned from the back side **406** by etching, mechanical grinding, CMP, the like, or any other suitable process. In some embodiments, the semiconductor substrate **317** is thinned to about 5 μm or less. In some embodiments, the semiconductor substrate **317** is thinned to about 3 μm or less. [0081] As shown by the cross-sectional view **3600** of FIG. **36**, a mask **3601** may be formed and used to etch trenches **3603** in the semiconductor substrate **317**. The trenches **3603** form a grid with segments between adjacent photodiodes **311A-D**. The trenches **3603** have a high aspect ratio. In some embodiments, the trenches **3603** have an aspect ratio of 15:1 or greater. In some embodiments, the trenches **3603** have an aspect ratio of 20:1 or greater. In some embodiments, the trenches **3603** have an aspect ratio of 25:1 or greater. In some embodiments, the trenches **3603** about the shallow trench isolation structures **333**. In the present example, a back side deep trench isolation structure is formed. Alternatively, a front side deep trench isolation structure may be formed.

[0082] Continuing with the present example, the trenches **3603** may be filled as shown by the cross-sectional view **3700** of FIG. **37** to provide the DTI structure **413**. In some embodiment, the trenches **3603** are filled with dielectric. In other embodiments, the trenches **3603** are lined with dielectric and then filled with conductive material to provide a conductive core. The conductive core may be grounded or may be coupled to a biasing voltage source.

[0083] The DTI structure **413** may comprise one or more dielectric layers. As these layers are deposited in the trenches **3603**, they also deposit on the back side **406**. Some of these layers may be thicker on the back side **406** depending on the condition the deposition processes with which they are formed.

[0084] In some embodiments, the trenches **3603** are lined with a high-K dielectric layer. The high-k dielectric passivates defects by forming an electric field that accumulates holes along the internal sidewalls, thereby passivating charge carriers (e.g., electrons). The high-K dielectric layer may be or comprise, for example, hafnium oxide (HfO_2), hafnium silicon oxide (HfSiO), hafnium silicon oxynitride (HfSiON), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), hafnium zirconium oxide (HfZrO), hafnium oxide aluminum oxide ($\text{HfO}_2\text{—Al}_2\text{O}_3$), zirconium oxide (ZrO_2), tantalum oxide (Ta_2O_5), aluminum oxide (Al_2O_3), yttrium oxide (Y_2O_3), lanthanum oxide (La_2O_3), strontium titanium oxide (SrTiO_3), or the like and may have a thickness in the range from 5 to 50 Angstroms, for example. The high-K dielectric layer may be deposited by ALD, CVD, PVD, the like, or any suitable process.

[0085] After lining, the trenches **3603** may be filled with an oxide such as silicon dioxide (SiO_2), tantalum pentoxide (Ta_2O_5), or the like. The fill may be deposited by ALD, CVD, PVD, the like, or any suitable process. In some embodiments, the oxide includes at least a layer of tantalum pentoxide or the like. Tantalum pentoxide has a refractive index between that of silicon (Si) and that of silicon dioxide. According, a layer of tantalum pentoxide can reduce reflections. The DTI structure **413** may be planarized on the back side **406**. Planarization may be by CMP, the like, or any other suitable process. Planarization provides a level surface on which to build the subsequent structure.

[0086] As shown by the cross-sectional view **3700** of FIG. **37**, additional dielectric layers and the back side metal grid **411** may be formed over the back side **406**. The back side metal grid **411** may have any suitable composition and thickness. The back side metal grid **411** may comprise aluminum (Al), copper (Cu), tungsten (W), the like, or any other suitable material.

[0087] As shown by the cross-sectional view **3900** of FIG. **39**, the color filters **409** and the microlenses **407** may then be formed. The color filters **409** may include red, green, and blue color filters. Other color combinations, such as cyan, yellow, and magenta, may be used instead. The color filters **409** may be in a Bayer pattern or some other pattern. The process may conclude with wafer dicing and packaging.

[0088] FIG. **40** provides a flow diagram for a process **4000** of forming an image sensor of the

present disclosure. While the process **4000** is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events is not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

[0089] The process **4000** begins with a group of acts **4001** which are performed on a first wafer and a group of acts **4021** which are performed on a second wafer. The acts **4001** may be performed before, after, or simultaneously with the group of acts **4021**.

[0090] Act **4003** is forming STI structures on the first wafer. The cross-sectional view **1400** of FIG. **14** provides an example. Act **4005** is implanting dopants to form photodiodes. The cross-sectional view **1500** of FIG. **15** provides an example. Forming the photodiodes may include a plurality of dopant implantations some of which may be carried out earlier or later in the sequence of acts **4001**. For example, photodiode formation may begin with a deep n-well implant made using high energy ions and without a mask.

[0091] Act **4007** is forming transistors on the front side. These include transfer gates, DCG gates, and source followers. The cross-sectional views **1600-1700** of FIGS. **16-17** provide an example. Acts **4009-4013** are back-end-of-line (BEOL) processing that forms a metal interconnect structure over a front side of the first wafer. The cross-sectional view **1800** of FIG. **18** provides an example. Act **4011** is an optional step of forming a LOFIC within the a metal interconnect structure. The cross-sectional views **2500-3000** of FIGS. **25-30** provide an example of how the LOFIC may be formed. Act **4015** is forming a bonding layer. The cross-sectional views **1900-2100** of FIGS. **19-21** and the plan view **2200** of FIG. **22** provide an example.

[0092] Act **4023** is forming STI structures on the second wafer. The cross-sectional view **2300** of FIG. **23** provides an example. Act **4025** is forming transistors on the second wafer. The transistor on the second wafer include at least row select transistors. Acts **4027** is forming the lower portion of a metal interconnect structure on the second wafer. The cross-sectional view **2400** of FIG. **24** provides an example. Act **4029** is an optional step of forming a LOFIC within the a metal interconnect structure. The cross-sectional views **2500-3000** of FIGS. **25-30** and the plan view **2200** of FIG. **22** provide an example of how the LOFIC may be formed. Acts **4031** is forming the upper portion of a metal interconnect structure on the second wafer. The cross-sectional view **3100** of FIG. **31** provides an example. Act **4033** is forming a bonding layer. The cross-sectional view **3200** of FIG. **32** provides an example.

[0093] Act **4040** is aligning the first and second wafers and bonding them together through their respective bonding layers. The cross-sectional view **3400** of FIG. **34A** provides an example.

[0094] Act **4041** is thinning the second wafer from the back side, forming TSVs through the second wafer's substrate, and forming contact pads on the back side. The cross-sectional view **3410** of FIG. **34B** provides an example. Act **4042** is bonding the second wafer to the third wafer so that the first, second, and third wafers are all bonded together. The cross-sectional view **3420** of FIG. **34C** provides an example.

[0095] Act **4043** is thinning the first wafer from the back side. The cross-sectional view **3500** of FIG. **35** provides an example. Act **4045** is etching deep trenches in the back side. The cross-sectional view **3600** of FIG. **36** provides an example. Act **4047** is filling the trenches to provide a DTI structure. The cross-sectional view **3700** of FIG. **37** provides an example. The trenches may be lined with one or more layers of high-K dielectric and then filled with another dielectric. Alternatively, the trenches may be filled with a conductive material after being lined with dielectric. Another alternative is to form a front side DTI structure rather than a back side DTI structure.

[0096] Act **4049** is forming a back side metal grid. The cross-sectional view **3800** of FIG. **38**. Act

4051 is additional processing that completes the formation of an image sensor. This may include forming color filters and microlenses. The cross-sectional view **3900** of FIG. **39** provides an example.

[0097] Some aspects of the present disclosure relate to an image sensor that a photodetector comprising a photosensitive area, a floating diffusion node, a transfer gate, a source follower, a row select transistor, a first dual conversion gain transistor, and a reset transistor. The photosensitive area, the transfer gate, the source follower, and the first dual conversion gain transistor are on a first die. The row select transistor is on a second die. The two dies are bonded together, and the source follower is connected to the row select transistor through the metal interconnect structures of the two dies.

[0098] In some embodiments, opening the first dual conversion gain transistor electrically isolates the floating diffusion node from the second die. In some embodiments, the photodetector further comprises a first LOFIC which is on the second die together with the reset transistor. In some embodiments, the photodetector further comprises a second LOFIC and a second dual conversion gain transistor. The second LOFIC is on the first die and the second dual conversion gain transistor is on the second die.

[0099] In some embodiments, the photodetector comprises two contact pads on the first die interfacing with two contact pads on the second die. In some embodiments, the two contact pads on the first die are in an array comprising active contact pads and dummy contact pads. The active contacts pads are in first rows, the dummy contact pads are in second rows, and the first rows are interleaved with the second rows.

[0100] In some embodiments, the first LOFIC is a three-dimensional metal-insulator-metal capacitor. In some embodiments, the first LOFIC is in horizontal alignment with a V.sub.dd rail in the second metal interconnect structure so that the first LOFIC is directly between the V.sub.dd rail and the second die.

[0101] In some embodiments, the photodetector further comprises an LOFIC on the first die coupled to the floating diffusion node through the first dual conversion gain transistor. In some embodiments, the photosensitive area is one of four photosensitive areas coupled to the floating diffusion node through four respective transfer gates. In some embodiments, the source follower and the first dual conversion gain transistor are aligned in a row, a first two of the four photosensitive areas are on one side of the row, and a second two of the four photosensitive areas are on an opposite side of the row. In some embodiments, a shallow trench isolation structure has segments between the row and a respective two of the four photosensitive areas on either side of the row. In some embodiments, there is a V.sub.dd rail on the first die extending parallel to the row. The V.sub.dd rail is connected to a drain of the source follower.

[0102] In some embodiments, the image sensor includes a back side deep trench isolation structure having segments surrounding the photosensitive area. In some embodiments, the floating diffusion node includes a wire in a first metallization layer on the first die, but the floating diffusion node is electrically isolated from higher metallization layers when the first dual conversion gain transistor is open.

[0103] Some aspects of the present disclosure relate to a photodetector including a photodiode in a first die, a floating diffusion node, a transfer gate between the floating diffusion node and the photodiode, and a plurality of additional transistors including a first transistor on the first die and a second transistor on a second die. The floating diffusion node is confined to the first die when the first transistor is open. In some embodiments, the photodetector includes an LOFIC on the second die and the floating diffusion node is coupled to the LOFIC through the first transistor.

[0104] Some aspects of the present disclosure relate to a method of making an image sensor. The method includes forming a photodiode and a floating diffusion region on a first wafer. A transfer gate, a source follower, and a first dual conversion gain transistor are also formed on the first wafer. The transfer gate is configured to selectively couple the photodiode to the floating diffusion region.

A first metal interconnect structure is formed over the first wafer. The first metal interconnect structure couples the floating diffusion region to a gate of the source follower and to a source side of the first dual conversion gain transistor. The method further includes forming a row select transistor on a second wafer and a second metal interconnect structure over the second wafer. The first wafer and the second wafer are bonded so that the first metal interconnect structure and the second metal interconnect structure are facing. The bonding couples a drain side of the row select transistor to a source side of the source follower. The first wafer is thinned and microlenses are formed on the first wafer. In some embodiments, the method further includes forming a first 3D-MIM capacitor in the second metal interconnect structure. One or more transistors, including the first dual conversion gain transistor, selectively couple the first 3D-MIM to the floating diffusion region. In some embodiments, a second dual conversion gain transistor is formed on the second wafer and the one or more transistors include the second dual conversion gain transistor and a second 3D-MIM capacitor is formed in the first metal interconnect structure. The first dual conversion gain transistor selectively couples the second 3D-MIM capacitor to the floating diffusion region.

[0105] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An image sensor, comprising: a first die comprising a first semiconductor substrate and a first metal interconnect structure; a second die comprising a second semiconductor substrate and a second metal interconnect structure, wherein the second die is bonded to the first die; and a photodetector comprising a photosensitive area, a floating diffusion node, a transfer gate, a source follower, a row select transistor, a first dual conversion gain transistor, and a reset transistor; wherein the photosensitive area, the transfer gate, the source follower, and the first dual conversion gain transistor are on the first die and the row select transistor is on the second die; and the source follower is connected to the row select transistor through the first metal interconnect structure and the second metal interconnect structure.
2. The image sensor of claim 1, wherein opening the first dual conversion gain transistor electrically isolates the floating diffusion node from the second die.
3. The image sensor of claim 1, wherein: the photodetector further comprises a first lateral overflow integration capacitor (LOFIC); and the first LOFIC and the reset transistor are on the second die.
4. The image sensor of claim 3, wherein: the photodetector further comprises a second LOFIC and a second dual conversion gain transistor; and the second LOFIC is on the first die and the second dual conversion gain transistor is on the second die.
5. The image sensor of claim 3, wherein the photodetector comprises two contact pads on the first die interfacing with two contact pads on the second die.
6. The image sensor of claim 5, wherein the two contact pads on the first die are in an array comprising active contact pads and dummy contact pads, wherein the active contacts pads are in first rows, the dummy contact pads are in second rows, and the first rows are interleaved with the second rows.
7. The image sensor of claim 3, wherein the first LOFIC is a three-dimensional metal-insulator-

metal capacitor.

8. The image sensor of claim 7, wherein the first LOFIC is in horizontal alignment with a V.sub.dd rail in the second metal interconnect structure so that the first LOFIC is between the V.sub.dd rail and the second die.

9. The image sensor of claim 1, wherein: the photodetector further comprises an LOFIC coupled to the floating diffusion node through the first dual conversion gain transistor; and the LOFIC is on the first die.

10. The image sensor of claim 1, wherein the photosensitive area is one of four photosensitive areas coupled to the floating diffusion node through four respective transfer gates.

11. The image sensor of claim 10, wherein the source follower and the first dual conversion gain transistor are aligned in a row, a first two of the four photosensitive areas are on one side of the row, and a second two of the four photosensitive areas are on an opposite side of the row.

12. The image sensor of claim 11, further comprising a shallow trench isolation structure comprising segments between the row and a respective two of the four photosensitive areas on either side of the row.

13. The image sensor of claim 11, further comprising a V.sub.dd rail on the first die extending parallel to the row, wherein the V.sub.dd rail is connected to a drain of the source follower.

14. The image sensor of claim 1, further comprising a back side deep trench isolation structure having segments surrounding the photosensitive area.

15. The image sensor of claim 1, wherein: the first metal interconnect structure includes a first metallization layer which is closest to the first die, and a second metallization layer, which is second closest to the first die; and the floating diffusion node includes a wire in the first metallization layer; and the floating diffusion node is electrically isolated from the second metallization layer when the first dual conversion gain transistor is open.

16. An image sensor, comprising: a first die comprising a first substrate and a first metal interconnect structure; a second die comprising a second substrate and a second metal interconnect structure, wherein the second die is bonded to the first die; and a photodetector comprising a photosensitive area within the first substrate, a floating diffusion node, a transfer gate between the floating diffusion node and the photosensitive area, a first transistor on the first die, and a second transistor on the second die; wherein the floating diffusion node is confined to the first die when the first transistor is open.

17. The image sensor of claim 16, further comprising an LOFIC on the second die, wherein the floating diffusion node is coupled to the LOFIC through the first transistor, the first metal interconnect structure, and the second metal interconnect structure.

18. A method of making an image sensor, the method comprising: providing a first substrate; forming a photodiode and a floating diffusion region in the first substrate; forming a transfer gate, a source follower, and a first dual conversion gain transistor on the first substrate, wherein the transfer gate is configured to selectively couple the photodiode to the floating diffusion region; forming a first metal interconnect structure over the first substrate, wherein the first metal interconnect structure couples the floating diffusion region to a gate of the source follower and to a source side of the first dual conversion gain transistor; providing a second substrate; forming a row select transistor on the second substrate; forming a second metal interconnect structure over the second substrate; and bonding the first substrate to the second substrate, wherein the bonding couples a drain side of the row select transistor to a source side of the source follower.

19. The method of claim 18, further comprising forming a first three-dimensional metal-insulator-metal capacitor in the second metal interconnect structure, wherein one or more transistors including the first dual conversion gain transistor selectively couple the first three-dimensional metal-insulator-metal capacitor to the floating diffusion region.

20. The method of claim 19, further comprising: forming a second dual conversion gain transistor on the second substrate, wherein the one or more transistors include the second dual conversion

gain transistor; and forming a second three-dimensional metal-insulator-metal capacitor in the first metal interconnect structure, wherein the first dual conversion gain transistor selectively couples the second three-dimensional metal-insulator-metal capacitor to the floating diffusion region.
